NETWORKED CONTROL

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EXERCIZES

1) Consider the following continuous-time system $\mathcal{S}$.

$$\dot{x}(t) = (\log_e 3)x(t) + (\log_e 9)u(t)$$
$$y(t) = x(t)$$

A) Assume that the system is controlled using a continuous-time control law of the type $u(t) = Ky(t)$. Define the range of values of K that make the closed-loop system asymptotically stable.

B) Compute the parameters $F, G, \text{and } H$ of the corresponding discrete-time dynamical equation

$$x_{k+1} = Fx_k + Gu_k, \qquad y_k = Hx_k$$

with sampling time $h = 1$ s.
Also, assuming that the system is controlled using a digital control law of the type $u_k = Ky_k$, define the range of values of K that make the closed-loop system asymptotically stable.

C) Assume that the system $\mathcal{S}$ is controlled through the networked control system (NCS) sketched in the following figure.

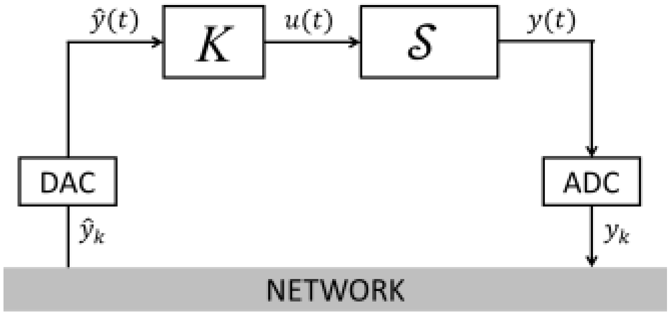

Figure 1: Networked control system

Answer the following questions considering **the case where** $K = -3/4$.

a. Define succinctly the **MATI (maximal allowable transfer interval) of a NCS**.
b. In case the delay is $\tau = 0$ s, compute the MATI of the NCS.
c. Assume now that $h = 1$ s, the NCS is subject to packet dropouts but the delay is absent, i.e., $\tau = 0$ s. Variable θ_k has the following periodic behavior.

k	1	2	3	4	5	6	7	8	9	$\cdots$
θ_k	0	0	1	0	0	1	0	0	1	$\cdots$

Is the corresponding NCS asymptotically stable?

d. Is the NCS asymptotically stable in case $h = 1$ s, the delay is $\tau = 0.5$ s, and in absence of packet dropouts?

SOLUTION

A) Setting $u(t) = Ky(t) = 3x(t)$, the closed loop dynamics is described by the equation

$$\dot{x}(t) = (\log_e 3)x(t) + 2(\log_e 3)u(t) = (1 + 2K)(\log_e 3)x(t)$$

whose eigenvalue is $\lambda = (1 + 2K)(\log_e 3)$. The system is closed-loop asymptotically stable for $K < -1/2$.

B) The corresponding discrete-time system is

$$\begin{cases} x_{k+1} & = & 3x_k + 4u_k \\ y_k & = & x_k \end{cases}$$

Setting $u_k = Ky_k = Kx_k$, the closed loop dynamics is described by the equation

$$x_{k+1} = (3 + 4K)x_k$$

whose eigenvalue is $\lambda = 3 + 4K$. The system is closed-loop asymptotically stable iff $|3 + 4| < 1$, i.e., iff $-1 < K < -1/2$.

C) See the following answers.

a. See the lecture notes.

b. In case $\tau = 0$, to compute the MATI we can consider the discretized scalar system $x_{k+1} = 3^h x_k + 2(3^h - 1)u_k$. Setting $u_k = -3/4x_k$, we obtain $x_{k+1} = \left(3^h - \frac{3}{2}(3^h - 1)\right)x_k = \left(\frac{3}{2} - \frac{1}{2}3^h\right)x_k$. To guarantee asymptotic stability, we need to compute the range of values of h guaranteeing that $\left|\frac{3}{2} - \frac{1}{2}3^h\right| < 1$, i.e., $0 < h < \log_3 5$. The MATI is then equal to $\log_3 5 = 1.465$ s.

c. To analyze the stability of the NCS under packet dropouts (but in absence of delays) defined by the sequence of values of variable θ_k, we first recall that the transition matrices in case transmission occurs and in case of dropout are, respectively

$$\widetilde{\Phi}_1 = \Phi(h,\tau) = \Phi(1,0) = \begin{bmatrix} 3 + 4K & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\widetilde{\Phi}_0 = \begin{bmatrix} 3 & 4\mathrm{K} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -3 \\ 0 & 1 \end{bmatrix}$$

Note, in passing, that $\widetilde{\Phi}_1$ has eigenvalues in $\lambda_1 = 0$ and $\lambda_2 = 0$, while $\widetilde{\Phi}_0$ has eigenvalues 3 and 1. Since the sequence of values of θ_k is 0,0,1,0,0,1,0,0, …:

The update rule that can be derived is, for all $k \in \mathbb{N}$,

$$z_{3k+3} = \widetilde{\Phi}_0\widetilde{\Phi}_0\widetilde{\Phi}_1 z_{3k}$$

$$z_{3k+4} = \widetilde{\Phi}_1\widetilde{\Phi}_0\widetilde{\Phi}_0 z_{3k+1}$$

$$z_{3k+5} = \widetilde{\Phi}_0\widetilde{\Phi}_1\widetilde{\Phi}_0 z_{3k+2}$$

Therefore, $\widetilde{\Phi}_1\widetilde{\Phi}_0\widetilde{\Phi}_0$, $\widetilde{\Phi}_0\widetilde{\Phi}_0\widetilde{\Phi}_1$ and $\widetilde{\Phi}_0\widetilde{\Phi}_1\widetilde{\Phi}_0$ can be regarded as the three-steps transition matrices for the NCS state. We resort to the fact that $\widetilde{\Phi}_1\widetilde{\Phi}_0\widetilde{\Phi}_0$, $\widetilde{\Phi}_0\widetilde{\Phi}_0\widetilde{\Phi}_1$ and $\widetilde{\Phi}_0\widetilde{\Phi}_1\widetilde{\Phi}_0$ have the same eigenvalues to study the convergence properties of the NCS. In particular

$$\widetilde{\Phi}_1\widetilde{\Phi}_0\widetilde{\Phi}_0 = \begin{bmatrix} 0 & 0 \\ 9 & -12 \end{bmatrix}$$

whose eigenvalues are -12 and 0. The NCS is therefore unstable.

d. In case packet dropouts are absent, but a delay is present, the transition matrix $\Phi(h,\tau)$ is

$$\Phi(h,\tau) = \begin{bmatrix} 3^h + \Gamma(h-\tau)2(\log_e 3)K & 3^{h-\tau}\Gamma(\tau)2(\log_e 3)K \\ 1 & 0 \end{bmatrix}$$

where $\Gamma(\tau) = \int_0^\tau e^{(\log_e 3)v}dv = \frac{1}{\log_e 3}(3^\tau - 1)$. In case $h = 1$ s, $\tau = 0.5$ s, and $K = -3/4$,

$$\Phi = \begin{bmatrix} -\frac{3}{2}(\sqrt{3}-3) & \frac{3}{2}(\sqrt{3}-3) \\ 1 & 0 \end{bmatrix}$$

whose eigenvalues are $\lambda_{1,2} = 0.9510 \pm 0.9988j$, whose modulus is equal to 1.3791>1. The NCS is therefore unstable.

2) Define, in a concise way, the different **components and sources of delay** in control networks.

SOLUTION: See the lecture notes.

3) When a **control network** is said to be **unstable**? *Hint: this question concerns stability of a control network, not stability of a networked control system.*

SOLUTION: See the lecture notes.